AI-Powered Logistics Disruption Monitoring System

(Monitoring real-world disruptions and identifying shipments at risk)

Team Name : ‘0’ and ‘1’
Hackathon : ZEN AI HACKATHON
Team Members :

R. Samyukta
S. Moses Gnanaraj
R. Gokul

PROBLEM STATEMENT :

Current Challenges of logistics :

Every week shipment routes are disrupted by:

- heavy rain
- road closures
- traffic congestion
- port delays

Major Issues

- disruptions detected too late
- shipments get delayed
- customers are informed late
- operational costs increase

There is **no intelligent system that automatically detects disruptions and identifies shipments at risk** and Everything is **monitored and handled manually by** the logistics team.

PROPOSED SOLUTION

Our Solution

We developed an **AI-powered logistics monitoring platform** that continuously analyzes real-world signals and predicts shipment risks.

Steps:

Automatically Collect disruption signals from weather, news, and traffic data using APIs

↓

Use AI to detect disruption events and determine severity

↓

Match detected disruptions with active shipment routes

↓

Identify shipments that are at risk of delay or disruption

↓

Predict delivery delays and suggest actions (rerouting, ETA update, customer notification)

Output :

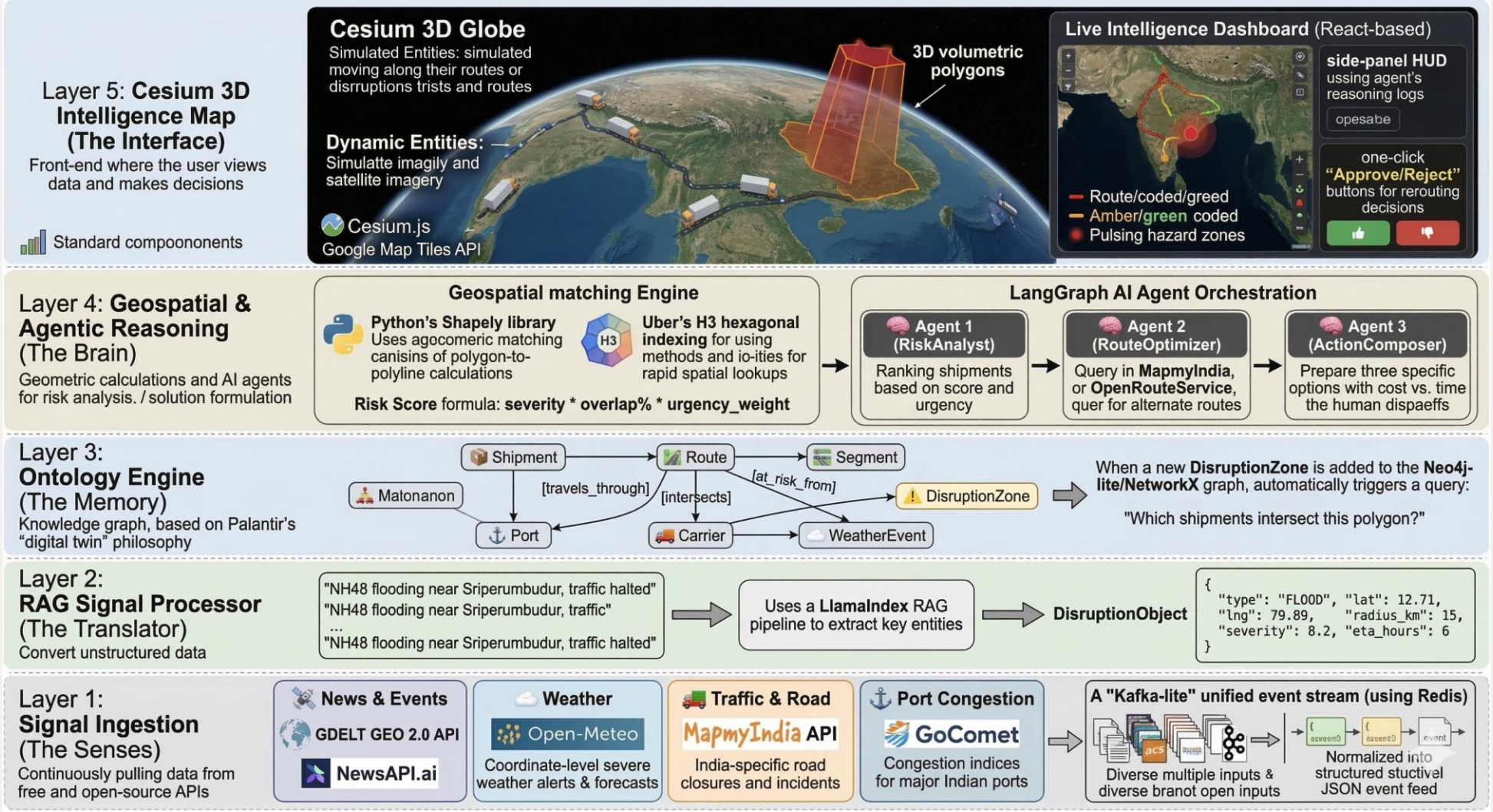
Automation in,

- risk alerts
- delay predictions
- alternative route suggestions
- customer notifications

UNIQUE FEATURES

1. Pre-Disruption Detection
2. Intelligent Risk Scoring
3. Delay Prediction (Estimated time of Arrival)
4. Cost Impact Estimation
5. AI Decision Engine
    - Rerouting shipments
    - Updating delivery ETA
    - Notifying customers

LAYERED ARCHITECTURE :



Tech Stack

- **Frontend / UI:** React in & next.js- Role-Based Dashboards
- **Backend Core:** Python & FastAPI - High-speed API bridge
- **AI Engine:** LangGraph & Groq (Llama-3) - 6-Agent reasoning pipeline
- **Routing :** CesiumJS - for 3D globe route and description rendering
- **Live Data:**
    - Open-Meteo - weather forecast
    - NewsAPI - news
    - GoComet - ports
    - Mappls / MapmyIndia - traffic
    - OSRM for route geometry fallback
    - OpenRouteService as route fallback
    - Synthetic scenarios for demo continuity when live APIs are absent

IMPACTS AND FUTURE ENHANCEMENTS :

Operational Impact :

- Early disruption detection
- Faster decision making
- Reduced delivery delays

Business Impact

- Optimized logistics routes
- Reduced operational costs
- Improved shipment reliability

Customer Impact

- Proactive delay notifications
- Better delivery transparency
- Improved customer satisfaction

Future Enhancements :

Live IoT & Telematics

- Right now, our AI looks at outside disruptions like weather and traffic. In the future, we will connect directly to IoT sensors inside the truck.

THANK YOU